Implementing grades in python Version 1.1 GPA CALCULATOR

```
courses = int(input("Enter number of courses: "))

data = []

for i in range(courses):

    print(f"\nCourse {i + 1}")

    course_name = input("Enter course name: ")
    credit_hours = float(input("Enter credit hours: "))
    obtained_marks = float(input("Enter obtained marks: "))
    total_marks = float(input("Enter total marks: "))

    data.append([
        course_name,
        credit_hours,
        obtained_marks,
        total_marks
    ])

def get_grade(percentage):

    if percentage >= 90:
        return "A", 4.0

    elif percentage >= 85:
        return "A-", 4.0

    elif percentage >= 80:
        return "B+", 3.3

    elif percentage >= 75:
        return "B", 3.0

    elif percentage >= 70:
        return "B-", 2.7

    elif percentage >= 65:
        return "C+", 2.3

    elif percentage >= 60:
        return "C", 2.0

    elif percentage >= 55:
        return "D", 1.0

    else:
        return "F", 0.0

total_points = 0
total_credits = 0
```

```
print("\n")
print("=" * 80)

print(
    f"{'Course':<20}"
    f"{'Credit':<10}"
    f"{'Obtained':<12}"
    f"{'Total':<10}"
    f"{'Percentage':<12}"
    f"{'Grade':<8}"
    f"{'Point':<8}"
)

print("=" * 80)


for course in data:

    course_name = course[0]
    credit_hours = course[1]
    obtained_marks = course[2]
    total_marks = course[3]

    percentage = (obtained_marks / total_marks) * 100

    grade, grade_point = get_grade(percentage)

    points = credit_hours * grade_point

    total_points += points
    total_credits += credit_hours

    print(
        f"{course_name:<20}"
        f"{credit_hours:<10}"
        f"{obtained_marks:<12}"
        f"{total_marks:<10}"
        f"{percentage:<12.2f}"
        f"{grade:<8}"
        f"{grade_point:<8}"
    )


print("=" * 80)

if total_credits > 0:

    gpa = total_points / total_credits

    print(f"\nTotal Credit Hours: {total_credits}")
    print(f"Your Final GPA is: {gpa:.2f}")

else:

    print("\nNo credit hours were entered.")
```

```
Enter number of courses: 5

Course 1
Enter course name: AI
Enter credit hours: 3
Enter obtained marks: 77
Enter total marks: 100

Course 2
Enter course name: DB
Enter credit hours: 4
Enter obtained marks: 80
Enter total marks: 100

Course 3
Enter course name: English
Enter credit hours: 3
Enter obtained marks: 67
Enter total marks: 100

Course 4
Enter course name: DATA Mining
Enter credit hours: 3
Enter obtained marks: 90
Enter total marks: 100

Course 5
Enter course name: DATA WAREHOUSE
Enter credit hours: 4
Enter obtained marks: 75
Enter total marks: 100


================================================================================
Course              Credit    Obtained    Total     Percentage  Grade   Point
================================================================================
AI                  3.0       77.0        100.0     77.00       B       3.0
DB                  4.0       80.0        100.0     80.00       B+      3.3
English             3.0       67.0        100.0     67.00       C+      2.3
DATA Mining         3.0       90.0        100.0     90.00       A       4.0
DATA WAREHOUSE      4.0       75.0        100.0     75.00       B       3.0
================================================================================

Total Credit Hours: 17.0
Your Final GPA is: 3.12
```